

Homework (Jalapeno)

- Q1.** In ancient Egypt, a goldsmith mixed $\frac{1}{4}$ kg of gold with $\frac{3}{4}$ kg of copper to create an alloy. What is the ratio of gold to the total mixture?
(A) 1:4
(B) 1:3
(C) 1:2
(D) 2:3
(E) 3:4
- Q2.** The ancient Greeks used a mixture of $\frac{2}{3}$ wine and $\frac{1}{3}$ water for a ceremonial drink. If 240 liters of the mixture is needed, how many liters of wine should be used?
(A) 120
(B) 160
(C) 180
(D) 200
(E) 220
- Q3.** A Roman merchant mixed $\frac{3}{5}$ kg of expensive spices with $\frac{2}{5}$ kg of cheaper spices. What is the ratio of expensive spices to cheaper spices?
(A) 1:2
(B) 2:3
(C) 3:2
(D) 3:1
(E) 2:1
- Q4.** In the construction of the Great Pyramid, workers used a mixture of $\frac{1}{2}$ limestone and $\frac{1}{2}$ granite for the outer casing. If 1000 blocks of the mixture are needed, how many blocks of limestone should be used?
(A) 250
(B) 333
(C) 400
(D) 500
- (E) 750
- Q5.** The ancient Chinese mixed $\frac{2}{5}$ kg of silk with $\frac{3}{5}$ kg of wool to create a fabric. What is the ratio of silk to wool?
(A) 1:2
(B) 1:1
(C) 2:3
(D) 2:1
(E) 3:2
- Q6.** A mixture of $\frac{3}{4}$ honey and $\frac{1}{4}$ water was used in ancient Greek medicine. If 480 grams of the mixture is needed, how many grams of honey should be used?
(A) 120
(B) 180
(C) 240
(D) 360
(E) 380
- Q7.** The ancient Egyptians used a mixture of $\frac{1}{3}$ myrrh and $\frac{2}{3}$ frankincense for ceremonial purposes. What is the ratio of myrrh to the total mixture?
(A) 1:3
(B) 1:2
(C) 1:1
(D) 2:3
(E) 3:2
- Q8.** A Roman recipe called for a mixture of $\frac{2}{3}$ olive oil and $\frac{1}{3}$ vinegar for a salad dressing. If 210 milliliters of the mixture is needed, how many milliliters of olive oil should be used?
(A) 70
(B) 100
(C) 120
(D) 140
(E) 160

Open Ended Questions (FRQ)

- Q9.** In ancient Mesopotamia, a mixture of clay and straw was used to build houses. If a mixture of $\frac{3}{5}$ clay and $\frac{2}{5}$ straw is used to make 150 bricks, how many bricks will contain $\frac{1}{2}$ kg of clay? Show your work and provide a clear answer.
- Q10.** The ancient Greeks used a mixture of $\frac{1}{4}$ gold and $\frac{3}{4}$ copper to create a statue. If the statue requires 40 kg of the mixture, and the price of gold is 50 per kg and the price of copper is 10 per kg, what is the total cost of the materials? Provide a step-by-step solution and a clear final answer.

Chef's Tips

- Provide clear instructions
- Allow the use of calculators
- Emphasize the importance of showing work